TATIANNA A. HERNUPONT

Software Engineering Intern | Junior Software Engineer | Robotics Software

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PROFESSIONAL SUMMARY

- Software engineer and robotics programmer with hands-on experience building autonomous and driver-control systems in C++ and Java.
- Strong foundation in debugging, version control (Git), and sensor integration (IMU, encoders) to produce reliable behavior.
- Seeking a Software Engineering Internship or Junior Software Engineer role focused on backend, embedded, robotics, or systems development.

TECHNICAL SKILLS

Languages: C++, Java, Python, Arduino (C-based)

Tools: Git, Visual Studio Code

Concepts: Object-Oriented Programming (OOP), Debugging

Robotics & Control Systems: PID Control, Odometry, IMU, Encoders, Tracking Wheels

Embedded Systems: Arduino Sensor Integration, Sensor Calibration, Real-Time Debugging

CAD Software: AutoCAD, SolidWorks, Revit, Fusion 360

Hardware Platforms: VEX V5, FIRST Robotics (FRC), BEST Robotics, Raspberry Pi

EXPERIENCE

FIRST Robotics Competition (FRC) — Lead Programmer — Dec 2025 – Present

Robotics Team

- Lead engineer of robot control development in Java, supporting drivetrain and mechanism control for competition tasks.
- Build and tune PID control loops for consistent performance across changing field conditions and battery levels.
- Use Git for team collaboration; and maintain a stable main branch.
- Coordinate integration testing with mechanical teams and execute structured troubleshooting during build sessions.
- Document configuration, tuning values, and operator controls to ensure reliable handoff.

BEST Robotics — Embedded Systems Programmer — Dec 2024 – Dec 2025

Competition Robotics Team

- Programmed Arduino-based sensor systems and integrated sensor feedback.
- Performed sensor calibration and validation to improve reliability.
- Assisted with troubleshooting wiring and software integration issues during testing and competition preparation.

VEX V5 Robotics — Lead Programmer (Nationals Qualifier) — 2024 – Present

Competition Robotics Team

- Develop and maintain autonomous routines and driver-control software in C++ using a structured codebase.
- Implement PID-based motion control (turning, driving, and alignment) to reduce overshoot.
- Integrate IMU, encoders, and tracking wheels to support accurate navigation.
- Debug software/hardware issues under pressure using logs, test runs, and tuning.
- Collaborate with mechanical teammates to align software behavior with physical constraints (gear ratios, friction, torque limits).
- Contributed to team success leading to qualification for VEX V5 Nationals.

PROJECTS

PID Tuning & Motion Control Toolkit — 2024 – Present

C++ / Java · Control Systems

- Developed a repeatable tuning workflow for PID constants, including step tests and controlled setpoint changes.
- Reduced drift and overshoot by refining control loop structure and adding safety limits to prevent unstable outputs.
- Improved maintainability by separating sensor reads and control logic.

Autonomous Navigation & Odometry Module (VEX / Robotics) — 2024 – Present

C++ · IMU · Encoders · Tracking Wheels

- Implemented odometry calculations using wheel encoders and tracking wheels to estimate robot position over time.
- Integrated IMU heading to stabilize angle estimation and improve autonomous path consistency.
- Created reusable movement primitives (drive-to-point, turn-to-heading, follow heading) to speed up autonomous development.

EDUCATION

Waltrip High School — Class of 2026

Houston, TX

- GPA: 3.85 — Top 15% of class.
- Relevant Coursework: AI & Cybersecurity (Python), Engineering Design, AP Pre-Calculus.

CERTIFICATIONS & PROGRAMS

- Certified SOLIDWORKS Associate (CSWA) (2026)
- AutoCAD Certified User (2025)
- Rice University STEM Program — Completion (2024)
- University of Houston–Downtown STEM Program — Completion (2022)